ComplianceKB: An Auditable Regulatory Knowledge Model for Human-Supervised Compliance AI Reasoning

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AI Agent / AI Infrastructure / AI Safety / Compliance Audit

Background and motivation

Compliance Audit / AI Infrastructure

A plant faces several rule systems at once

Four OSHA frameworks can apply to the same plant: PSM, HazCom, Recordkeeping, and LOTO. Auditors must locate each regulatory requirement (what the employer must do) and preserve its interpretation.



Industrial context only, not an evaluated workplace.

The missing operational layer

To our knowledge, open, source-linked requirement collections with reviewed operational labels remain scarce. Rule languages and regulatory RAG provide foundations, but auditors still need a maintained link from source text to proposed checks.

A relevant paragraph does not explicitly preserve every applicability assumption, activating event, evidence category, or review decision. A maintained record layer makes those interpretations reusable and inspectable rather than recreating them silently for each query.

Retrieving a clause is only the first step: auditors must identify when it applies and what evidence to inspect. ComplianceKB links these reviewed interpretations to their sources, supporting traceable AI proposals and human review.

Can a reusable, source-linked obligation representation make operational compliance outputs easier to reproduce and inspect, and which judgments still depend on people?

We specify RKM and implement it in ComplianceKB.

Regulatory Knowledge Object

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$$RKO = \langle r, A, P, O \rangle$$

r Requirement frame
ID, requirement, modality, risk

A Applicability
Activity, jurisdiction, industry

P Provenance
Clause, dated excerpt, hash, review

O Operational semantics
Trigger, evidence, review method

Grouped manuscript Eq. (1): r = (id, u, m), O = (t, e, v). Applicability remains descriptive, not a complete legal predicate.

v1.2.0: 108 records. PSM 55; HazCom 16; Recordkeeping 26; LOTO 11. Types: 103 obligations, 3 prohibitions, 2 permissions.

Status, curator, and review date support traceable correction. JSON / CSV / JSON-LD / SQL exports retain consistent IDs.

Worked source-to-audit example

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OSHA-PSM-018: 29 CFR 1910.119(g)(3)

“The employer shall prepare a record which contains the identity of the employee, the date of training, and the means used to verify that the employee understood the training.”

Field	Released value / interpretation
Requirement	Ascertain and document receipt and understanding of required process-operator training.
Evidence / method	training_record / record_review
Trigger	onboarding_new_employee Additional: training_refresh
Provenance	2024-07-01; normalized SHA-256 Prefix: 9d3264fe53abd1d4

Inspect employee identity, training date, and the means of verifying understanding. The citation supports the requirement; the workplace record supports its performance.

OSHA-HAZCOM-001, §1910.1200(e)(1): written hazard communication program. Trigger = always_on; evidence = written_program.

What this work contributes

AI Infrastructure / AI Agent / Human-AI Verification

1 A formal record model

Each regulatory requirement states what an employer must do. Its scope, source, trigger, and evidence link that requirement to an audit action.

2 A traceable curation workflow

Official clause → AI draft → human review → frozen release → cited output. Inspectable artifacts preserve each interpretation; this is a proposed method, not an adopted standard.

3 A controlled representation comparison

Across five inputs, RKM has the highest trigger and evidence F1 for both models. The same hybrid candidate IDs support the paired comparison shown at right.

4 Verification with visible limitations

Different checks expose different problems. Review of the original 89 records: 22 PASS, 6 PASS_WITH_NOTE, 61 FAIL_NEEDS_CORRECTION.

5 A defined role for human judgment

AI proposes checks; people verify sources, workplace evidence, and final judgments. Added RKM fields also add tokens; shared-label development scenarios have no independent holdout.

Human work and limits

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AI retrieves records and drafts source-linked audit proposals.

People check site facts, exceptions, authentic evidence, and final compliance judgments.

Unmeasured: labor savings, calibration, transfer, temporal reasoning. Hashes detect excerpt drift only.

Inputs and curation workflow

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Excerpt-only input	Excerpt + RKM input
Same stored excerpt: employee identity, training date, and means used to verify understanding.	Same excerpt + applicability and provenance; trigger: onboarding_new_employee; evidence: training_record.

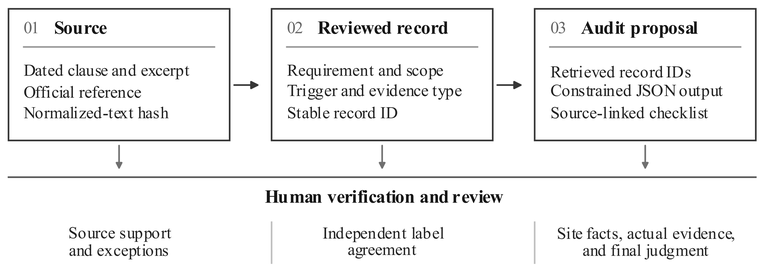


Figure 1. Source checks precede reusable records and audit proposals.

Benchmark method and reproducibility

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24 scenarios × 5 conditions × 2 models = 240 calls.

Five inputs: all excerpts, BM25, MiniLM, hybrid, and RKM.
One model call returns cited IDs, labels, and a decision in JSON.

$$RRF(d) = \sum_s [60 + rank_s(d)]^{-1}$$

Top 20 per retriever, top 10 after fusion; absent ranks contribute zero. BM25 k1=1.5, b=0.75. MiniLM: L2-normalized exact dot product.

$$F_1(G, P) = \frac{2\,|G \cap P|}{|G| + |P|}$$

DEV-001 GPT hybrid: 4 reference IDs, 5 predicted, 3 overlaps. F1 = 2×3/(4+5) = 0.667; average across cases.

Paired 24-row bootstrap: 10,000 resamples, seed 20260927, 95% percentile intervals for RKM-minus-hybrid differences.

gpt-5.6-terra / claude-sonnet-5, low reasoning, 2,000-token cap; Python 3.12.14, PyTorch 2.14.0+cpu, sentence-transformers 6.1.0, all-MiniLM-L6-v2, BM25. Retained outputs permit API-free scoring.

Paired operational-label results

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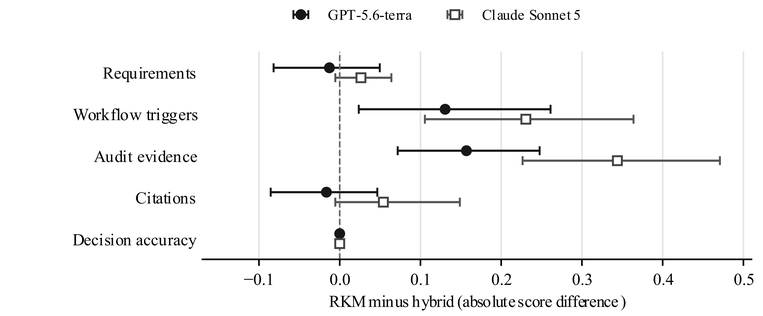


Figure 2. Positive values favor RKM over hybrid; 95% paired intervals.

RKM improves recovery of when to check (triggers) and what to inspect (evidence). Both models' paired intervals exclude zero for these labels.
No consistent requirement/citation gain; unchanged decisions. Label recovery is not legal correctness.

Model and retrieval comparisons

AI Agent / AI Infrastructure

F1 measures overlap with reference labels; higher is better. Decision is exact-match accuracy. RKM leads on triggers and evidence.

Model / context	Req. F1	Trigger F1	Evidence F1	Decision
GPT / all excerpts	.881	.690	.606	.917
GPT / BM25	.825	.710	.716	.958
GPT / MiniLM	.800	.707	.677	.917
GPT / hybrid	.801	.693	.681	.917
GPT / RKM	**.789**	**.824**	**.838**	**.917**
Claude / all excerpts	.819	.535	.554	.958
Claude / BM25	.768	.568	.530	.958
Claude / MiniLM	.681	.614	.502	.917
Claude / hybrid	.745	.628	.544	.958
Claude / RKM	**.771**	**.858**	**.888**	**.958**

Recall@10 measures reference requirements recovered in ten results; MRR and nDCG reward earlier relevant hits. Hybrid has the highest recall; E5 is post-hoc.

Retriever	Recall@10	MRR@10	nDCG@10
BM25	.804	.847	.754
MiniLM	.801	.765	.691
BM25 + MiniLM	.817	.835	.749
E5 (post-hoc)	.784	.847	.730
BM25 + E5 (post-hoc)	.808	.858	.753

Layers of verification

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Different layers test different claims. Annotator agreement measures consistency, not legal truth.

Layer / completed evidence	Can show / cannot establish
Schema and regression tests	Valid structure and exports; not legal fidelity.
Provision-presence check	Represented topics; not exhaustive requirement coverage.
Author source review: 108/108	Source support with AI assistance; not independent expert correctness.
Blinded second annotation: 50 IDs	Nominal κ: risk .234, evidence .642, trigger .443. Agreement is not truth.
Controlled benchmark: 24 / 240	Operational-label recovery; not independent deployment performance.

References

Selected sources for regulations, related work, and retrieval:

[1] OSHA. 29 CFR §§1910.119, 1910.1200, 1910.147; Part 1904.

[2] Sun, Luo & Li (2025). A Compliance Checking Framework Based on Retrieval Augmented Generation. COLING, 2603–2615.

[3] Reimers & Gurevych (2019). Sentence-BERT. EMNLP-IJCNLP, 3982–3992.

[4] Cormack, Clarke & Buettcher (2009). Reciprocal rank fusion. SIGIR, 758–759.

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